PRATHAM KUMAR

AI / GenAI Engineer — LLM Evaluation, Agentic Systems, RAG

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SUMMARY

AI Engineer with 1+ year at Google (via Highspring) and 3 prior AI/ML internships. Built LLM evaluation frameworks (1,000+ responses/day, ~85% F1, ~90% manual QA automated), ML + LLM pipelines across ~5M entities, and RAG/agentic systems in LangChain and LangGraph.

TECHNICAL SKILLS

Languages & Data: Python, SQL, MongoDB, C/C++, Power BI
AI/ML: PyTorch, TensorFlow, Scikit-learn, Transformers, OpenCV, NumPy, Pandas
Agentic & GenAI: LangChain, LangGraph, RAG, LLM Evaluation, LangSmith, Prompt Engineering, Hugging Face, MCP, N8N
Development & Cloud: FastAPI, Flask, REST API, Docker, AWS, GCP, Gradio, Streamlit, Databricks, Git, Linux,

EXPERIENCE

Associate Analyst (AI/ML) | Highspring (On-site, Google) *Jun 2025 – Present*

Gurugram | Python, Scikit-learn, LangChain, LLM Evaluation, SQL, Knowledge Graph, GCP

- Engineered a hybrid KG + ML + LLM Wikipedia analysis pipeline (scraping, NLP extraction, Gemini 3.5 Flash) across 10 locales and 5 verticals, covering ~5M entities. Compressed a months-long manual content-gap process to hours (~40% cost reduction).
- Designed a LightGBM content-richness classifier (Good/Average/Poor) to predict Comprehensiveness and Readability, trained on 400+ labelled articles with per-vertical models and locale-calibrated thresholds. Benchmarked Wikipedia depth against KG entries and found 29.3% of entities were richer on Wikipedia, quantifying the business case for acquisition.
- Built an LLM Evaluation framework scoring 1,000+ LLM responses/day at ~85% F1 score across Correctness, consistency, Relevance & Completeness, automating ~90% of manual QA effort and driving Gemini model/prompt iteration decisions.
- Validated 70+ upstream data-provider pipelines across 1.4M+ entities; found 17 providers drove ~63.8% of impact giving a clear prioritization signal for provider investment..

Data Science Intern | Haqdarshak *Feb 2025 – Jun 2025*

Delhi | Python, LangChain, FastAPI, AWS

- **Built and shipped a video-generation pipeline** that produced 350+ regional-language videos, cutting manual production effort by 70%.
- **Built a scalable REST API** to automate script generation and database updates, streamlining content workflows for the team.
- **Partnered with cross-functional teams** to align solutions with business goals, contributing to a 40% increase in regional user engagement.

Backend Engineer Intern | Verbaflo.AI *Sep 2024 – Dec 2024*

Gurugram | Python, FastAPI, MongoDB, Docker

- **Built audit logging** for database operations and system tracking to support compliance and debugging.
- **Engineered JWT-based OAuth 2.0 authentication** and end-to-end CRM workflows from scratch, enabling secure role-based access.
- **Developed 10+ REST APIs** for a Property Management System, improving data retrieval efficiency by 35%.

Machine Learning Intern | DraftAi *Jul 2024 – Aug 2024*

Delhi | Python, LangChain, Flask

- **Applied advanced RAG techniques** to legal document generation, achieving a 62.5% improvement in legal citation accuracy over GPT-4o.
- **Fine-tuned LLMs for domain-specific legal workflows,** reducing hallucinations in generated legal documents.
- **Scaled the pipeline to process 1,000+ legal pages** at 90%+ relevance, overcoming context-window limits through chunking and retrieval design.

PROJECTS

Secure RAG with Role-Based Access Control (RBAC) | *GitHub*

Python, FastAPI, ChromaDB, Ollama, Streamlit

- **Built an end-to-end RAG assistant (FastAPI + ChromaDB + Streamlit)** enforcing role-based access control at the vector-store level — junior users cannot retrieve C-suite documents, even via prompt injection.
- **Designed a 4-tier clearance + multi-tenant access model** with JWT AuthN/AuthZ, prompt-injection guardrails, and full audit logging — validated with an automated test suite proving cross-tenant and cross-clearance isolation.

EDUCATION

Guru Gobind Singh Indraprastha University, Delhi *Aug 2021 – May 2025*

B.Tech Artificial Intelligence and Data Science — CGPA: 8.59